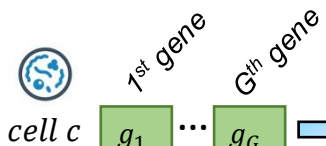


CellRefine Post-pretraining

Single-cell
foundational
model

CellRefine
Post-pretraining

Single Cell Data



Tokenizer
 $\mathcal{T}(g)$

Single-cell
foundational
model

Cell encoding

Gaussian
Mixture
Variational
Latent Layer

$\mathcal{L}_{GMVE}$

$\mathcal{L}_{MLM}$

z_c

Cell ontology-based
similarity regularization

$\mathcal{L}_{lineage}$

$\langle z_c, z_\alpha \rangle$

$\langle z_c, z_\beta \rangle$

$\vdots$

$\langle z_c, z_\gamma \rangle$

Cell-prototype
Matching

$\langle z_c, z_m^1 \rangle$

$\langle z_c, z_m^2 \rangle$

$\vdots$

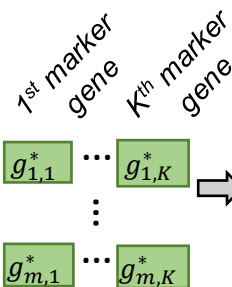
$\langle z_c, z_m^l \rangle$

$\mathcal{L}_{prototype}$

Marker Gene Data



Marker
organizer
 $H(c)$



Tokenizer
 $\mathcal{T}(g)$

Prototype
encoding

BERT
encoder
 (f_{Marker})

z_m^1

z_m^2

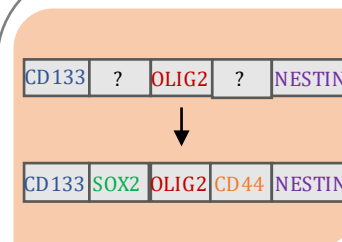
$\vdots$

z_m^l

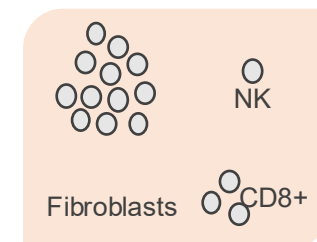
$$\text{Step 1: } \mathcal{L}_{Total} = \mathcal{L}_{MLM} + \lambda_1 \mathcal{L}_{prototype} + \lambda_2 \mathcal{L}_{lineage} + \lambda_3 \mathcal{L}_{GMVE}$$

Downstream application

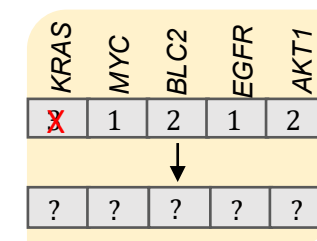
Fine-tuning on
downstream
tasks



Spatial
Transcriptomic
Imputation



Cell Identity
Prediction



Gene
Perturbation
Response
Prediction

$$\text{Step 2: } \mathcal{L}_{task}$$